

Ordered simplex caps and the three-observation Stringer bound

Richie Sater

Independent researcher, Silver Spring, Maryland, USA

richiesater@gmail.com

September 2026

Abstract

We prove that a cap of the uniform distribution on a tetrahedron, cut through an ordered barycentric point, has maximal volume at a step normal within the matching order cone. The proof reduces stationary caps to a section-centroid inequality, whose only nonsimplicial case is established by exact rational computation. Applied to the binomial Stringer bound for three i.i.d. observations in $[0, 1]$, the theorem gives distribution-free coverage at every confidence of at least $66.6314788\dots\%$. This range is sharp for point-wise domination of Gaffke’s valid mean bound. In every sample size, a two-knot perturbation gives an explicit obstruction to that comparison; its critical tail levels converge to $0.2846681370\dots$.

Keywords: Stringer bound; bounded mean; simplex cap; Dirichlet average.

1 Introduction

The Stringer bound [12] estimates the mean of a distribution on $[0, 1]$ from independent observations. In monetary-unit sampling these observations are taints, the fractions by which sampled account items are overstated. For a sample arranged as $t_{(1)} \geq \dots \geq t_{(n)}$, the binomial-factor bound at confidence $1 - \alpha$, where $0 < \alpha < 1$, is

$$\text{SB} = p_n(0) + \sum_{j=1}^n \{p_n(j) - p_n(j-1)\}t_{(j)}, \quad (1)$$

where, for $j < n$, the Clopper–Pearson factor $p_n(j)$ is the unique solution in $(0, 1)$ of

$$\sum_{i=0}^j \binom{n}{i} p^i (1-p)^{n-i} = \alpha, \quad p_n(n) = 1.$$

Its finite-sample conservatism asks whether

$$\mathbb{P}_F\{\text{SB} \geq \mu(F)\} \geq 1 - \alpha \quad (2)$$

for every distribution F on $[0, 1]$, with mean $\mu(F)$. The general question at conventional confidence levels remains open. Bickel [2] established asymptotic conservatism; Pap and van Zuijlen [10] identified the asymptotic confidence threshold of $1/2$, and their finite-sample

examples [9] show that low-confidence failure is possible. Bernoulli taints reduce the bound to Clopper–Pearson [3]. The present paper concerns independent draws, not systematic or other dependent sampling designs.

Our main result is geometric. Write $I_z(a, b)$ for the distribution function at z of a $\text{Beta}(a, b)$ random variable. A $\text{Dirichlet}(1, \dots, 1)$ vector is uniform on the simplex $\{d_i \geq 0 : \sum_i d_i = 1\}$.

Theorem 1 (Four-coordinate monotone cap theorem). *Let $D \sim \text{Dirichlet}(1, 1, 1, 1)$, let $c_0 \leq c_1 \leq c_2 \leq c_3$ be nonnegative with $\sum_i c_i = 1$, and let $x_0 \leq x_1 \leq x_2 \leq x_3$. Put $C_r = \sum_{i < r} c_i$. Then*

$$\Pr\{x \cdot D > c \cdot x\} \leq \max_{r=1,2,3} I_{C_r}(r, 4-r). \quad (3)$$

Moreover, the maximum over all nonconstant ordered x equals the right side: each candidate is attained by the step vector with its first r coordinates zero and its remaining coordinates one.

Thus a cap cut through the barycentric point c is largest when the normal x takes only two values. The assertion holds for every ordered c , not only for confidence-bound coefficients.

For observations $t_1, \dots, t_n \in [0, 1]$, define

$$G_\alpha(t) = Q_{1-\alpha} \left(D_0 + \sum_{i=1}^n t_i D_i \right), \quad D \sim \text{Dirichlet}(1, \dots, 1), \quad (4)$$

where $Q_q(X) = \inf\{z : \mathbb{P}(X \leq z) \geq q\}$. Gaffke’s mean test [4], proved valid by Vlassis and Thomas [11, Theorem 1], gives $\mathbb{P}_F\{G_\alpha \geq \mu(F)\} \geq 1 - \alpha$. The quantile formulation follows by test inversion [6, Theorem 5.1 and Corollary 5.2]. The geometric theorem yields the following exact comparison range.

Corollary 2 (Three-observation Stringer range). *Set*

$$\alpha_\star = \left(\frac{19 + \sqrt{21}}{34} \right)^3 = 0.3336852118672717\dots \quad (5)$$

For $n = 3$, the binomial Stringer bound dominates G_α on every sample if and only if $0 < \alpha \leq \alpha_\star$. Consequently (2) holds for every F at every nominal confidence of at least 66.6314788...%.

The endpoint concerns pointwise comparison, not the coverage threshold of Stringer itself. Proposition 6 gives the corresponding obstruction in every dimension.

The obstacle is that a simplex cap moves both its normal and its threshold: ordinary stochastic ordering of Dirichlet averages does not directly compare these probabilities. The Dirichlet-average and spline formulas are classical [1, 5]. The centroid case is related to the simplex halfspace-depth calculation of Paidaveine and Van Bever [8]; here the barycentric point is arbitrary and the normal is restricted to its order cone. The correspondence between stationary cuts and section centroids [7, Proposition 1.11] is also classical. These facts calculate a cap and locate its stationary points, but do not identify its global maximum in an order cone.

The key additional inequality bounds a cap by a beta probability formed from its section centroid. On a face where knots coincide, stationarity identifies the centroid's block masses with those of c . It is therefore essential to use prefixes ending at strict knot gaps. The resulting local inequality bounds every face maximum by a step cap. For a tetrahedron, the only nonelementary section has two knots on each side of the cut.

In that case, centroid order provides a polynomial constraint, and a nonnegative Bernstein multiplier proves the required inequality. A separate identity treats the repeated-upper-knot boundary. The finite calculation uses exact rational arithmetic; the stationary-face reduction, the Clopper–Pearson weight inequalities, and the limiting obstruction are ordinary proofs.

2 Section centroids and tetrahedral caps

For $N \geq 2$, let $D \sim \text{Dirichlet}(1, \dots, 1)$, let $x_0 \leq \dots \leq x_{N-1}$, put $T = x \cdot D$, and consider a regular section $T = q$. Write

$$p = \Pr(T > q), \quad \gamma_i = \mathbb{E}(D_i \mid T = q), \quad M_r = \sum_{i < r} \gamma_i. \quad (6)$$

At the step vector with its first r knots zero and its remaining knots one, the cap at the barycentric threshold $c \cdot x$ has value $I_{C_r}(r, N - r)$, where $C_r = \sum_{i < r} c_i$.

Lemma 3 (Section-centroid maximum principle). *Suppose that, in dimension N , every regular ordered section with $\gamma_0 \leq \dots \leq \gamma_{N-1}$ satisfies*

$$p \leq \max_{r: x_{r-1} < x_r} I_{M_r}(r, N - r). \quad (7)$$

Assume the same inequality on singular sections by continuity. Then, for every nondecreasing probability vector $c = (c_0, \dots, c_{N-1})$, every ordered x , and $C_r = \sum_{i < r} c_i$,

$$\Pr\{x \cdot D > c \cdot x\} \leq \max_{1 \leq r \leq N} I_{C_r}(r, N - r). \quad (8)$$

Proof. For constant x the cap is empty. Otherwise, first take every $c_i > 0$; then $q = c \cdot x$ lies strictly between the endpoint knots. Translation and positive scaling of x leave the cap unchanged, so normalize its least and greatest distinct knots to zero and one. On any face of this compact ordered-knot polytope, collect the equal consecutive knots into blocks. If their distinct values are $y_1 < \dots < y_k$, put $W_b = \sum_{i \text{ in block } b} D_i$ and let A_b be the corresponding sum of the c_i . Differentiation of the cap at a regular section gives

$$\frac{\partial}{\partial y_b} \Pr\left\{\sum_a y_a W_a > \sum_a y_a A_a\right\} = f(q)\{\mathbb{E}(W_b \mid T = q) - A_b\}, \quad (9)$$

where f is the density of T . At an interior knot level this identity extends by continuity of the section integrals.

At a relative-interior stationary point, the right side vanishes for every nonendpoint block. The two identities

$$\sum_b \{\mathbb{E}(W_b \mid T = q) - A_b\} = 0, \quad \sum_b y_b \{\mathbb{E}(W_b \mid T = q) - A_b\} = 0$$

then give the same conclusion for the endpoint blocks. Conditional symmetry within a block makes each γ_i its block mass divided by the block size. These values are consecutive block averages of the nondecreasing vector c , and hence are nondecreasing. Moreover, at every strict knot gap, $M_r = C_r$. The local barrier (7) therefore bounds the value at every relative-interior stationary point by the right side of (8). A nonstationary face maximum lies on a proper face; induction over the faces ends at a step vector, where the cap equals $I_{C_r}(r, N-r)$. Approximation and continuity cover the singular and nonstrict cases. $\square$

The first section with a nonsimplicial cross-section occurs when $N = 4$ and two knots lie on either side of the threshold. The next proposition is the local barrier needed in Lemma 3.

Proposition 4 (Tetrahedral section barrier). *Let $N = 4$ and use the notation in (6). If the section centroid is ordered, then*

$$p \leq \max_{r: x_{r-1} < x_r} I_{M_r}(r, 4-r). \quad (10)$$

In the two-versus-two crossing $x_0 < x_1 < q < x_2 < x_3$, the stronger inequality

$$p \leq I_{M_2}(2, 2) = 3M_2^2 - 2M_2^3$$

holds.

Proof. There are three crossing types. If only x_3 is above q , set $a_i = (x_3 - q)/(x_3 - x_i)$ for $i < 3$. The cap is a tetrahedron and its section is its base, so

$$p = a_0 a_1 a_2, \quad M_3 = (a_0 + a_1 + a_2)/3.$$

Thus $p \leq M_3^3 = I_{M_3}(3, 1)$ by AM–GM. If only x_0 is below q , set $b_i = (q - x_0)/(x_i - x_0)$ for $i > 0$. Then

$$p = 1 - b_1 b_2 b_3, \quad \gamma_0 = 1 - (b_1 + b_2 + b_3)/3, \quad \gamma_i = b_i/3.$$

The knot order gives $b_1 \geq b_2 \geq b_3$, while the centroid order gives the reverse inequalities. Hence all three equal a common b , and $p = 1 - b^3 = I_{M_1}(1, 3)$.

It remains to treat two knots on each side. Translation and scaling put the centered knots in the form

$$(-1, -u, tv, t), \quad 0 < u, v < 1, \quad t > 0. \quad (11)$$

For positive temporary variables A, B, C, D , the divided-difference formula [1, 5] and its derivative give

$$p(A, B, C, D) = \frac{C^3}{(C+A)(C+B)(C-D)} + \frac{D^3}{(D+A)(D+B)(D-C)}, \quad (12)$$

$$\gamma = f(0)^{-1}(-p_A, -p_B, p_C, p_D), \quad (13)$$

where the density $f(0)$ is obtained from (12) by replacing C^3, D^3 with $3C^2, 3D^2$. This is followed by $(A, B, C, D) = (1, u, tv, t)$. Exact simplification of the upper centroid-order condition gives

$$\gamma_3 - \gamma_2 = \frac{(v-1)R(u, v, t)}{\Delta_0(u, v, t)}, \quad \Delta_0 > 0, \quad (14)$$

where

$$R = t^4 v^2 (u^2 + u + 1) + 2t^3 uv(u + 1)(v + 1) + t^2 u^2 (v^2 + 4v + 1) - u^3.$$

Since $v < 1$, orderedness gives $R \leq 0$. The last positive term in R also gives $t^2(v^2 + 4v + 1) \leq u < 1$, so (11) lies in the unit cube.

With $M = \gamma_0 + \gamma_1$, exact rational simplification gives

$$3M^2 - 2M^3 - p = \frac{t^2 P(u, v, t)}{\Delta(u, v, t)}, \quad (15)$$

where P has tensor degree $(9, 9, 11)$ and

$$\Delta = 27(t + 1)^3(t + u)^3(tv + 1)^3(tv + u)^3(tuv + tv + uv + u)^3 > 0.$$

The remaining inequality follows from an exact Bernstein-multiplier identity. Recall that the tensor products of $B_i^d(z) = \binom{d}{i} z^i (1 - z)^{d-i}$ are nonnegative on a unit cube; thus nonnegative coefficients in this basis imply polynomial nonnegativity. There is a polynomial S of tensor degree $(6, 7, 7)$ with 79 positive Bernstein coefficients and all others zero. Exact integer and rational calculation shows that

$$H = P + RS$$

has 631 positive Bernstein coefficients of degree $(9, 9, 11)$ and all 569 remaining coefficients zero. The rational coefficients of S , and the integer-arithmetic derivation and sign calculation, are supplied with the code cited below. Hence on $R \leq 0$, $P = H - RS \geq 0$, proving (15).

When $v = 1$, equality of the upper knots removes the information in (14). The remaining centroid inequality has positive denominator and numerator

$$G = -2t^3 u - t^3 + t^2 u^2 - 4t^2 u + 3tu^2 + 3u^2.$$

Orderedness gives $G \geq 0$. If $t \geq 1$, then

$$G \leq (u - 1)\{(3u + 1)t + 4u\} \leq 0,$$

so the nondegenerate ordered domain again has $t < 1$. After the positive factor $t^2(1 - u)^2$ is removed from the comparison numerator, the boundary certificate checks the exact identity

$$P_{\partial} = H_{\partial} + 2G(1 - u)^2 t^2,$$

where all 30 degree- $(4, 5)$ Bernstein coefficients of H_{∂} are nonnegative (19 are positive, 11 are zero). This closes the repeated-upper boundary. The identity for $v < 1$ remains valid at $u = 1$, so repeated lower knots cause no further case. If $u = v = 1$, there are two knot blocks and the cap equals $I_{M_2}(2, 2)$. In the mixed crossing $r = 2$ is always a strict gap; in the endpoint crossings the active gap is respectively $r = 3$ or $r = 1$. For an interior knot level $q = x_2 < x_3$, the one-above formulas still hold with $a_2 = 1$. If $x_0 < q = x_1 < x_2$, the section is a triangle with $\gamma_1 = 1/3$ and $\gamma_2 = (q - x_0)/(3(x_2 - x_0)) < 1/3$, contrary to centroid order. These exhaust the interior knot levels, including $x_1 = x_2$ in the first case. $\square$

Proof of Theorem 1. Proposition 4 proves the hypothesis of Lemma 3 in every crossing stratum for $N = 4$, yielding (3). At the stated step vector, $\sum_{i < r} D_i \sim \text{Beta}(r, 4 - r)$ and the cap event is $\sum_{i < r} D_i < C_r$, which proves attainment. $\square$

3 The sharp Stringer comparison

Affine normalization turns comparison with (4) into the cap problem above. The normalization is useful for any sample size.

Proposition 5 (Simplex-cap criterion). *Arrange a sample as $t_1 \leq \dots \leq t_n$. If $t_1 < 1$, put*

$$z_0 = 0, \quad z_k = \frac{t_{k+1} - t_1}{1 - t_1} \quad (1 \leq k \leq n-1), \quad z_n = 1,$$

and

$$s_{n,\alpha}(z) = p_n(0) + \sum_{k=1}^{n-1} (p_n(n-k) - p_n(n-k-1)) z_k.$$

Then $\text{SB} \geq G_\alpha(t_1, \dots, t_n)$ if and only if

$$\Pr_D \left\{ \sum_{k=1}^{n-1} z_k D_k + D_n > s_{n,\alpha}(z) \right\} \leq \alpha, \quad 0 \leq z_1 \leq \dots \leq z_{n-1} \leq 1. \quad (16)$$

Consequently, proving (16) on the complete ordered knot simplex proves distribution-free Stringer coverage at (n, α) .

Proof. Rewrite (1) for ascending observations and subtract t_1 from both the Stringer threshold and the $n+1$ knots $(t_1, \dots, t_n, 1)$ in the Dirichlet average. Division by $1 - t_1$ gives the displayed threshold and knots. Exchangeability of the Dirichlet weights permits the stated labeling. Because the normalized average has knots zero and one, its distribution is continuous, so its $(1 - \alpha)$ -quantile is at most the threshold exactly when its strict upper tail there is at most α . If $t_1 = 1$, every observation and both bounds equal one. The coverage conclusion follows from the validity of G_α . $\square$

Proof of Corollary 2. Write $x = p_3(0)$, $y = p_3(1)$, and $z = p_3(2)$. In the ascending-knot form of Proposition 5, Stringer's threshold has weight vector

$$c = (1 - z, z - y, y - x, x).$$

At the three step vectors, the beta-binomial identity makes the corresponding cap probability exactly α . Theorem 1 therefore proves the cap criterion once c is nondecreasing.

Put

$$f(s) = 1 - 3s^2 + 2s^3, \quad f(y) = \alpha, \quad x = 1 - \alpha^{1/3}, \quad z = (1 - \alpha)^{1/3}.$$

The function f is strictly decreasing on $(0, 1)$. The last weight inequality is $y \leq 2x$. It is automatic if $2x \geq 1$; otherwise it is equivalent to

$$\alpha - f(2x) = x(-3 + 15x - 17x^2) \geq 0. \quad (17)$$

The first root of the quadratic factor is $x_\star = (15 - \sqrt{21})/34$, and $\alpha = (1 - x)^3$, so the last weight inequality holds exactly through $\alpha_\star$.

For the middle inequality, put $r = \alpha^{1/3} = 1 - x$. Since $\alpha \leq \alpha_\star < 1/2$, we have $z \geq r$ and $z^3 + r^3 = 1$, while

$$4 \left\{ f\left(\frac{x+z}{2}\right) - \alpha \right\} = 3(z-r)(z^2 + r^2 - 1) \geq 0.$$

Thus $y \geq (x+z)/2$. Finally, $z > 3/4$ and

$$\alpha - f(2z-1) = (1-z)(17z^2 - 19z + 5) > 0,$$

so $y \leq 2z - 1$. These are precisely the three inequalities saying that c is nondecreasing. The cap criterion and finite-sample validity of G_α prove the first assertion.

For sharpness, suppose $\alpha > \alpha_\star$. Then $x < x_\star < 1/2$, and (17) has the opposite strict sign; since f is decreasing, $y > 2x$. Thus the last two threshold weights satisfy $c_2 = y - x > c_3 = x$. Consider the ordered knot family $k(h) = (0, 0, h, 1)$ and write

$$\Phi(h) = \Pr\{k(h) \cdot D > c \cdot k(h)\}.$$

At $h = 1$, the beta-binomial identity gives $\Phi(1) = \alpha$. Conditional on $D_2 + D_3 = c_2 + c_3$, symmetry gives $\mathbb{E}(D_2 \mid D_2 + D_3 = c_2 + c_3) = (c_2 + c_3)/2$. The derivative formula (9) therefore yields

$$\Phi'(1) = \frac{g_\alpha}{2}(c_3 - c_2) < 0,$$

where $g_\alpha > 0$ is the density of $D_2 + D_3$ at $c_2 + c_3$. Consequently $\Phi(h) > \alpha$ for $h < 1$ sufficiently close to one, and Proposition 5 gives $\text{SB} < G_\alpha$ on the corresponding sample. $\square$

4 A terminal-edge obstruction in every dimension

The same perturbation identifies a necessary condition for the comparison at every sample size.

Proposition 6 (Terminal-edge phase transition). *For each $n \geq 2$, let r_n be the unique solution in $(1/2, 1 - 1/(2n))$ of*

$$(2r-1)^{n-1}\{2n-1-2(n-1)r\} = r^n, \tag{18}$$

and put $\alpha_n = r_n^n$. If $\alpha > \alpha_n$, then some ordered sample satisfies $\text{SB} < G_\alpha$. Moreover,

$$\alpha_n \longrightarrow \alpha_\infty = 0.28466813704083846168\dots, \quad \alpha_\infty(1 - 2\log \alpha_\infty) = 1. \tag{19}$$

Thus for every $\alpha > \alpha_\infty$, pointwise Stringer–Gaffke domination fails for all sufficiently large n .

Proof. Write $p_0 = p_n(0)$ and $p_1 = p_n(1)$, and consider $x(\varepsilon) = (0^{n-1}, 1 - \varepsilon, 1)$. If $D \sim \text{Dirichlet}(1, \dots, 1)$, then

$$S = D_{n-1} + D_n \sim \text{Beta}(2, n-1), \quad W = D_{n-1}/S \sim \text{Unif}(0, 1)$$

independently. The random score is $S(1 - \varepsilon W)$ and its barycentric threshold is $p_1 - \varepsilon(p_1 - p_0)$. Differentiating the conditional beta survival probability at zero gives

$$\left. \frac{d}{d\varepsilon} \Phi_c(x(\varepsilon)) \right|_{\varepsilon=0} = f_{2,n-1}(p_1) \left(\frac{p_1}{2} - p_0 \right). \quad (20)$$

Here $\Phi_c(x) = \mathbb{P}\{x \cdot D > c \cdot x\}$, and $f_{2,n-1}$ is the beta density. The cap at zero is the calibrated step cap of probability α . Consequently it rises above α exactly when $p_1 > 2p_0$; equivalently, the last two threshold weights reverse order because $c_{n-1} = p_1 - p_0 > p_0 = c_n$. The cap criterion then gives $\text{SB} < G_\alpha$ for the corresponding sample.

Put $r = \alpha^{1/n}$, so $p_0 = 1 - r$. Substituting $p_1 = 2p_0$ in its defining binomial tail gives (18). For

$$H_n(r) = \frac{(2r - 1)^{n-1} \{2n - 1 - 2(n - 1)r\}}{r^n},$$

direct differentiation yields

$$\frac{d}{dr} \log H_n(r) = \frac{n(2nr - 2n + 1)}{r(2r - 1)(2nr - 2n - 2r + 1)}.$$

The last denominator factor is negative. Hence H_n increases to $r = 1 - 1/(2n)$ and then decreases strictly to $H_n(1) = 1$, proving the unique nontrivial crossing and the stated direction of its sign. Finally, for fixed α ,

$$H_n(\alpha^{1/n}) \longrightarrow \alpha(1 - 2 \log \alpha).$$

The limiting function has exactly one nontrivial root in $(0, 1)$; the convergence is locally uniform around that root, so the unique finite roots converge to it. This proves (19) and the eventual-failure assertion. $\square$

For $n = 3$, this critical level is exactly α_* . The derivative identifies why comparison fails: the last two barycentric weights reverse order. It does not construct an i.i.d. distribution with Stringer undercoverage. The tetrahedral theorem therefore suggests two distinct questions: extending the active-gap section inequality to higher dimensions, and proving coverage beyond the range of pointwise comparison.

Data and code availability

The exact rational multiplier, the boundary identity, and their derivation and sign checks are available in the accompanying source repository.

AI disclosure

Anthropic Claude and OpenAI Codex were used for mathematical exploration, code development, literature searches, and drafting; the author accepts responsibility for the manuscript. The author received no external funding and declares no competing interests.

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